Reference Study	Input	Output	Dataset	Evaluation Metrics	Results	Positives	Negatives
Kucharski, Idzikowski & Sarnicka (2026) Benchmarking KWS on Google Speech Commands with a New Data Extension — Int. J. Electronics & Telecommunications, 72(3)	One-second, 16 kHz mono WAV keyword utterances (ITU-T P.800-compliant recording)	Keyword class label (35-word vocabulary; implicit silence/unknown classes)	Google Speech Commands v0.02 + Wroc■aw extension: 3,710 recordings from 106 non-native speakers (one recording per word per speaker)	Precision, Recall, F1-score, confusion matrices	>99% on both corpora; **CRNN-L 0.945/0.976**; plain **DNN** drops ≈28 points on new speakers (**0.65**); **Attention RNN** degrades to **78.45%** on original GSC	Rigorous reproducibility; evaluation on freshly recorded, unseen speakers as a faithful generalization measure	One recording per word per speaker (empty silence/unknown cells); apparent copy error in reporting
Sukhadia & Chowdhury (2026) HArnESS: Lightweight Arabic SSL Foundation Models via Iterative Distillation — SPEAKABLE@LREC 2026	Raw 16 kHz speech waveforms (Arabic and English)	ASR, Arabic dialect ID (ADI-5), and emotion recognition (KSU Emotion)	≈4,001 h Arabic + 3,565 h English pre-training (QASR, MGB3, LibriSpeech, CommonVoice, GigaSpeech, YouTube); evaluation on MGB2/MGB3, ADI-5, KSU Emotion	WER (ASR); classification accuracy (DID, SER)	**HArnESS-L**: WER **15.5%** (MGB2), **41.6%** (MGB3); SER **94.66%**; DID **84.98%** — outperforms HuBERT-L and XLS-R on all Arabic tasks; 79.4% compression costs +4.7 WER, −14.4 DID points	First family of Arabic SSL foundation models trained from scratch; principled accuracy–efficiency trade-off study	Frozen-encoder evaluation only (full fine-tuning may differ); indirect comparisons; Figure 2 plots WER on an inverted axis
Gan & Li (2025) SynTTS-Commands: Synthetic Speech Command Dataset for TinyML Keyword Spotting — arXiv:2511.07821	Synthetic one-second speech command utterances (WAV, 35.76 GB)	Command text (23 EN, 25 ZH) paired with speaker embeddings	384,621 synthesized clips (111.33 h) over 48 commands; timbres from VoxCeleb1/2 (5,994 speakers) and FreeST Mandarin (855); 70/15/15 speaker-disjoint split	Quality score Q = 100×(0.5·D/5 + 0.5·S/30); Paraformer ASR verification; accuracy; efficiency	**DS-CNN** with 30K params: **99.46% (EN) / 97.18% (ZH)**; **CRNN 99.50% / 97.42%**; INT8 **EfficientNet-B0** 97.93%	Large-scale, quality-filtered synthetic corpus; TTS augmentation shown viable for tiny-footprint KWS training	Train and test on the same generator (no real-speech evaluation); no OOV negatives or noise; English/Chinese only (no Arabic)
Cohn et al. (2026) Speech Recognition Challenges for Older Adults with Cognitive Impairment — CHI 2026	Read-aloud audio of 30 Alexa-style commands from older adults	ASR transcripts; per-segment WER statistically linked to acoustic-prosodic features	Virtual Assistant Session (VAS) subset of DementiaBank: 83 participants (30 cognitively normal, 8 MCI, 45 dementia)	Word Error Rate (WER); lme4 linear mixed-effects models, AICc comparison (MuMIn), VIF diagnostics	Mean WER = **0.32 (dementia)** vs ≈0.12 (controls), p<0.001; no significant normal–MCI gap (≈0.14); worst on IoT commands; **Whisper medium** helps dementia group most; errors correlate with higher shimmer, lower intensity, lower silence ratio	First systematic quantitative evidence of ASR inequity by cognitive status; actionable accessibility recommendations (delayed endpointing, button alternatives, caregiver-in-the-loop review)	Read (non-spontaneous) speech — conservative estimate; US English only; MCI classification based on MoCA score, not clinical diagnosis
Abdoli et al. (2026) Benchmarking Commercial ASR Systems on Code-Switched Speech (Arabic, Persian, German) — arXiv:2605.19069 (PerleAI)	Code-switched speech recordings (intra-sentential language mixing)	System rankings (ASR transcripts per API)	1,200 professionally recorded clips (300 per pair): Egyptian-English, Saudi-English, Persian-English, German-English (quiet environment, headset/USB mics)	WER (jiwer); multilingual BERTScore; Kendall's τ rank correlation	**ElevenLabs Scribe v2** leads: **13.2% WER** / **0.936 BERTScore** (OpenAI 38.6%, Google 39.4%, Azure 43.6%; Deepgram 5.0% German-only); rankings agree perfectly (τ=1.0); WER inflates inter-system gaps ≈3× vs BERTScore	First benchmark addressing Arabic/Persian code-switching; validates BERTScore as a semantically fairer metric; directly relevant to Arabic-English command evaluation	Commercial conflict of interest; scripted read-aloud speech; no confidence intervals or significance tests; modest per-pair sample; curated hard samples limit representativeness
Warlewski et al. (2026) SM-STMC: Streaming Keyword Spotting with Statically-Scheduled Short-Term Memory Convolutions — arXiv:2609.35005 (Samsung R&D;, Poland)	Streaming 16 kHz audio → 40-bin log-Mel spectrogram (window 1024, hop 256); utterances padded ±0.5 s	Keyword posteriors at each streaming step under real-time constraints	Google Speech Commands (10 commands + negative class) and an extended variant with 0.5 s silence appended per side	Accuracy, F1-score, MCPS, memory footprint (bytes), latency	**SM-STMC-1s**: **0.9382 acc / 0.9403 F1** vs **VGG1** 0.9278/0.9292; on 2 s padded input **0.9715 F1** vs 0.5798 (single-window VGG1); up to **–82% MCPS, –65% memory** (9,280 vs 17,184 B); real-time operation (every 4–8 frames)	Sustained streaming KWS with state-of-the-art efficiency on MCU-class hardware; no retraining required; deterministic static scheduling; fits our always-listening scenario	Evaluated only on the 10-command GSC task, not Arabic commands; headline figures correspond to recall; limited comparator set